



**AKADEMIA GÓRNICZO-HUTNICZA IM. STANISŁAWA STASZICA W KRAKOWIE**  
**Faculty of Electrical Engineering, Automatics, Computer Science and Biomedical Engineering**

Bachelor of Science Thesis

*Embedded Implementation and Comparison of Attitude Estimation  
Methods*

*Implementacja na urządzenie wbudowane algorytmów odtwarzania orientacji i ich  
porównanie*

|                   |                                |
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| Author:           | <i>Piotr Woliński</i>          |
| Degree programme: | <i>Automatics and Robotics</i> |
| Supervisor:       | <i>dr inż. Dawid Knapik</i>    |

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# 1. Abstract

The main goal of this thesis is to compare various attitude estimation algorithms (with a North unreferenced Yaw) using data from a 6-degree-of-freedom MEMS inertial measurement unit (6DOF IMU). The comparison will be made based on two main criteria, estimation accuracy and computation cost.

The experimental evaluation will include following algorithms:

- Complementary filter
- Mahony filter
- Extended Kalman Filter (EKF)
- Madgwick filter

The estimation accuracy will be measured under different conditions and at different IMU sampling frequencies. This data will be interpreted using the mentioned algorithms and compared with a reference attitude measurement (obtained by a reference sensor or a motion capture system). Test scenarios will cover measurements in both static conditions (with low external interference) and dynamic conditions (with linear or centrifugal accelerations added). The tests should enable comparison of how different algorithms handle noise directly from sensors and how those filters react to external noise, introduced by the environment.

To compare the computational cost, selected algorithms will be implemented and benchmarked on an STM32 family microcontroller in the C programming language. This will enable a direct measurement of time or memory overhead introduced by the filter.



## 2. Methods of Attitude Representation

An attitude of a body is a description of angular position of its body (sensor) frame, as a relation to the world (Earth) reference frame.

The most commonly known method for expressing the attitude of a body is through the use of Euler angles. This method describes the attitude as a sequence of three rotations around three separate axis of reference. While this method is simple and very intuitive it is also, arguably the most problematic of all methods mentioned here. The biggest problem of Euler angles representation is the phenomenon of gimbal lock. It appears when two rotation axis become parallel, which results in the loss of one degree of freedom. This can lead to singularities in the mathematical representation, which in turn causes difficulties and discontinuities in attitude representation. On top of that, calculating rotations of vectors between two frames described with Euler angles involves the use of, computationally expensive trigonometric functions. Additionally, due to the fact that rotations around each axis are calculated separately, the order of rotations matters, and gives different result based on chosen convention.

Most of the problems of Euler angles can be solved by using rotation matrices. A rotation matrix is a  $3 \times 3$  orthogonal matrix with determinant equal to 1, which represents a rotation in three-dimensional space. This method does not suffer from gimbal lock and provides a unique way of representing rotations, at a cost of reducing intuitiveness of the representation. Rotating a vector using a rotation matrix requires only matrix multiplication. The biggest issue with rotation matrices is their memory footprint. They require 9 parameters to represent a rotation, which is more than necessary and leads to additional numeric complexity and instability. The matrixes will also accumulate numerical errors during rotation concatenation, and will require re-orthogonalization to maintain their properties.

Because of that one more improvement can be made, when it comes to attitude representation, by using unit quaternions. Quaternions are a mathematical construct that extends complex numbers and can be used to represent rotations in three-dimensional space. It consists of four components: one real (scalar) part, and three imaginary parts forming a 3d vector. Quaternions bring the same improvements as rotation matrices, while also reducing the memory footprint to less than half. They also reduce the number of multiplications required for vector rotation and concatenation, which makes them a natural choice for representing rotations. Quaternions still suffer from the same numerical issues as rotation matrixes but maintaining the unit norm requires much less computational effort.

## 2.1. Representing Rotations as Quaternions

Quaternions are defined as a four-dimensional extension of complex numbers, consisting of one real part and three imaginary parts. A quaternion can be represented as:

$$\mathbf{q} = a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}, \quad \mathbf{q} \in \mathbb{H} \quad (2.1)$$

where  $a, b, c, d$  are real numbers and  $i, j, k$  are the basis vectors. The set of all quaternions is denoted by  $\mathbb{H}$ , for William Rowan Hamilton, who first introduced quaternions in 1843.

To simplify the notation, from now on quaternions will be represented as a 4D vector, where the first component is the scalar part and the last three components are the vector part:

$$\mathbf{q} = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \quad (2.2)$$

Multiplication between quaternions, called a Hamilton product is defined as:

$$\mathbf{q}_1 \mathbf{q}_2 = \begin{pmatrix} a_1 a_2 - b_1 b_2 - c_1 c_2 - d_1 d_2 \\ a_1 b_2 + b_1 a_2 + c_1 d_2 - d_1 c_2 \\ a_1 c_2 - b_1 c_2 + c_1 a_2 + d_1 b_2 \\ a_1 d_2 + b_1 c_2 - c_1 b_2 + d_1 a_2 \end{pmatrix} \quad (2.3)$$

Quaternion conjugate is defined as:

$$\mathbf{q}^* = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \quad (2.4)$$

For a quaternion  $\mathbf{q}$  to represent a valid rotation, it must have a unit norm:

$$\|\mathbf{q}\| = \sqrt{\mathbf{q}^* \mathbf{q}} = \sqrt{\mathbf{q} \mathbf{q}^*} = \sqrt{a^2 + b^2 + c^2 + d^2} = 1 \quad (2.5)$$

Let a quaternion representing a rotation of frame A with respect to frame B be denoted as  ${}^B_A \mathbf{q}$ , and a vector written in frame B be denoted as  ${}^B \mathbf{v}$ , with  $\mathbf{u}$  being a unit vector. Any rotation  $\theta$  around a unit vector  ${}^B \mathbf{u}$  can be represented as a quaternion:

$${}^B_A \mathbf{q} = \begin{pmatrix} \cos \frac{\theta}{2} \\ {}^B \mathbf{u} \sin \frac{\theta}{2} \end{pmatrix} \quad (2.6)$$



## 2.2. Chosen frame of reference

The global frame of reference will be represented using NED (North-East-Down) coordinate system. The NED system is a right-handed local tangent plane coordinate system defined. The  $x$ -axis of this system points towards the north, the  $y$ -axis points towards the east and the  $z$ -axis points downwards, aligning with local gravity vector.



### 3. Attitude from IMU Measurements

A typical MEMS IMU consists of 2 separate sensors. A rate gyroscope, measuring angular velocity along 3 sensor frame axis. As well as a 3-axis accelerometer, measuring linear acceleration along the same three axis. Neither one of those two sensors can be used alone to provide an reliable attitude observation in real conditions.

#### 3.1. Attitude from Accelerometer

Accelerometer readings  $y_a$  can be represented as:

$$\mathbf{y}_a = {}^W_B \mathbf{R}^W \mathbf{g} + {}^B \mathbf{a} + \mathbf{b}_a + \mathbf{w}_a \quad (3.1)$$

Where:

${}^W_B \mathbf{R}$  – Rotations matrix representing transformation from world to body frame

${}^W \mathbf{g}$  – Gravity vector in word frame

${}^B \mathbf{a}$  – Local linear acceleration of the sensor

$\mathbf{b}_a$  – Bias of the accelerometer

$\mathbf{w}_a$  – Measurement noise, including all unmodeled dynamics of the system

Assuming perfectly static conditions and 0 bias term, the accelerometer measures only the gravitational acceleration vector in the sensor frame. This measurement can be used to calculate a transformation aligning the body frame to the NED frame down direction. It is important to note that this method doesn't provide full orientation of the body, only the attitude can be estimated, while the heading (yaw) remains unobservable.

In most common cases, the sensor will be subject to additional, and often unpredictable external forces, which will make accelerometer only attitude estimation unreliable or even dangerous when used for control purposes.

#### 3.2. Attitude from Gyroscope

Gyroscope measurements  $y_g$  can be represented as:

$$\mathbf{y}_g = {}^B\boldsymbol{\omega} + \mathbf{b}_g + \mathbf{w}_g \quad (3.2)$$

Where:

${}^B\boldsymbol{\omega}$  – Angular velocity vector in the sensor frame

$\mathbf{b}_g$  – Gyroscope bias, slowly varying over time

$\mathbf{w}_g$  – Measurement noise, including all unmodeled dynamics of the system

Assuming an initial orientation, which cannot be measured by the gyroscope, the orientation can be estimated by integrating the angular velocity over time. This method gives an full orientation estimate, which is mostly unaffected by external linear accelerations. However, due to measurement noise, numerical integration errors and most importantly, the slowly varying bias, this estimate will drift over time, making it unreliable for long term use.

### 3.3. Attitude from Sensor Fusion

While achieving a reliable and universal attitude estimation from either of the two sensors alone is not possible, combining the measurements from both sensors can provide a much more accurate and stable estimate. The gyroscope measurement is much more reliable and accurate in short term observations, so its going to serve a the "base" of the algorithm. The accelerometer measurement will correct the errors that accumulate in the gyroscope based estimate, as well as providing an initial orientation estimate.

## **4. Chosen Attitude Estimation Algorithms**

### **4.1. Mahony filter**

Mahony filter is a name commonly used to refer to a group of complementary filters derived by Robert Mahony in [1]. The main idea behind those algorithms is to provide a computationally inexpensive solution for estimating attitude from cheap MEMS IMU sensors. Mahony expands on the idea of the classic complementary filter, by introducing a feedback term to the gyroscope measurement, which is proportional to the error between the accelerometer measurement and the expected gravity vector in the sensor frame. This feedback term helps to correct the drift in the gyroscope based estimate, while still maintaining the computational simplicity of a complementary filter.

In this work, the *explicit complementary filter* version of the algorithm proposed in [1] will be used.

### **4.2. AQUA Complementary filter**

### **4.3. Madgwick filter**

### **4.4. Extended Kalman Filter**



## Bibliografia

- [1] Robert Mahony, Tarek Hamel i Jean-Michel Pflimlin. „*Nonlinear Complementary Filters on the Special Orthogonal Group*”. W: *IEEE Transactions on Automatic Control* 53.5 (2008), s. 1203–1218. DOI: [10.1109/TAC.2008.923738](https://doi.org/10.1109/TAC.2008.923738).